

1)

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
    }
    else if (pid == 0)
    {
        // Child process
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        // Only runs if exec fails
        printf("Command execution failed!\n");
    }
    else
    {
        // Parent process
        printf("Parent PID: %d\n", getpid());

        wait(NULL);

        printf("Child process finished.\n");
    }
}
```

```

nithya@WIN-482M0QMA856:~$ nano practical1.c
nithya@WIN-482M0QMA856:~$ gcc practical1.c
nithya@WIN-482M0QMA856:~$ ./a.out
Enter a Linux command: ls
Parent PID: 1867
Child PID: 1868
Parent PID: 1867
OS_PRACTICE  b2      dirz      f8      open3.c      practice1.c
a.out        backup  exec1.c   linux_task1 open4.c      practice2.c
a1           bio.txt exec2.c   linux_task2 permission_lab skill1.c
a10          data.txt exec11.c  linux_task3 pipe         skill2.c
a10.c        dir1    f1        linux_task4 pipe1.c      skill3.c
a7           dir2    f2        mybio.txt  pipe2.c      square.c
a8           dir3    f3        numbers.txt pipe3.c      student.txt
a9           dir4    f4        open1      pipe4.c      student_portal
abc          dir5    f5        open1.c    pipe5.c      xyz
b1           dir6    f7        open2.c    practical1.c z7
Child process finished.

```

Linux terminal commands

```

nithya@WIN-482M0QMA856:~$ uname
Linux
nithya@WIN-482M0QMA856:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               48 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:        0-15
Vendor ID:                   AuthenticAMD
Model name:                  AMD Ryzen 7 7730U with Radeon Graphics
CPU family:                  25
Model:                      80
Thread(s) per core:         2
Core(s) per socket:         8
Socket(s):                   1
Stepping:                    0
BogoMIPS:                    3992.39
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge m
ca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall
nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep
_good nopl xtopology tsc_reliable nonstop_tsc cpuid ex
td_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16
sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand h
ypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a
misalignsse 3dnowprefetch osvw topoext perfctr_core ss
bd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb sha_ni
xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat
npt nrip_save tsc_scale vmcb_clean flushbyasid decodea
ssists pausefilter pfthreshold v_vmsave_vmload umip va
es vpclmulqdq rdpid fsrm
Virtualization features:
Virtualization:              AMD-V
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                         256 KiB (8 instances)
L1i:                         256 KiB (8 instances)
L2:                           4 MiB (8 instances)
L3:                          16 MiB (1 instance)
NUMA:
NUMA node(s):                1
NUMA node0 CPU(s):          0-15

```

```
nithya@WIN-482M0QMA856:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/

nithya@WIN-482M0QMA856:~$ ps
  PID TTY          TIME CMD
  364 pts/0    00:00:00 bash
 1980 pts/0    00:00:00 ps

nithya@WIN-482M0QMA856:~$ top
top - 13:33:22 up 59 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  25 total,   1 running, 24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 6849.6 total, 6095.2 free,  609.1 used,  299.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,   0.0 used,  6240.5 avail Mem

   PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24284   15372  11516 S   0.0   0.2   0:01.89 systemd
    2 root        20   0    3180    2212   2072 S   0.0   0.0   0:00.02 init-syste+
    6 root        20   0    3180    2040   1940 S   0.0   0.0   0:00.00 init
   52 root        19  -1   50400   16764  15464 S   0.0   0.2   0:00.54 systemd-jo+
   69 systemd+    20   0   22416   14444  11944 S   0.0   0.2   0:00.18 systemd-re+
   91 root        20   0   35260   12288   9176 S   0.0   0.2   0:03.17 systemd-ud+
```

Command

Role/Purpose

uname Displays the os being used.

lscpu Displays CPU architecture, cores, threads, and processor information.

lsblk Displays storage devices, partitions, and their sizes.

ps Displays currently running processes and their PIDs.

top Provides real-time information about processes and CPU/memory usage.

2)

```
#include<stdio.h>
#include<unistd.h>
#include<fcntl.h>
int main()
{
int fd1;
int fd2;
char buf[100];
size_t n;
fd1=open("a7", O_RDWR);
fd2=open("z7", O_RDWR);
printf("Files successfully opened and created.\n");
printf("%d\n",fd1);
printf("%d\n",fd2);
n=read(fd1,buf,30);
//printf("%s\n",buf);
write(fd2,buf,n);
}
```

```
nithya@WIN-482M0QMA856:~$ nano open3.c
nithya@WIN-482M0QMA856:~$ gcc open3.c
nithya@WIN-482M0QMA856:~$ ./a.out
Files successfully opened and created.
3
4
nithya@WIN-482M0QMA856:~$ |
```

Strace:

```
nithya@WIN-482M0QMA856:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffdbd7bc4d8 /* 25 vars */) = 0
brk(NULL)                               = 0x57f2a9be9000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7eb370055000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27889, ...}) = 0
mmap(NULL, 27889, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7eb37004e000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7eb37001b000
mmap(0x7eb370022000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x7eb370022000
mmap(0x7eb370043000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7eb370043000
mmap(0x7eb37004a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x7eb37004a000
mmap(0x7eb37004c000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7eb37004c000
```

3)

```
#include<unistd.h>
#include<stdio.h>
int main(){
int pid;
pid=fork();
if(pid==0){
printf("Child Process\n");
printf("Child process id is %d\n",getpid());
}
else if(pid>0){
printf("Parent Process\n");
printf("Parent process id is %d\n",getppid());
}
else if(pid<0)
{
printf("Failed process\n");}
}
```

```

nithya@WIN-482M0QMA856:~$ nano practical2.c
nithya@WIN-482M0QMA856:~$ gcc practical2.c
nithya@WIN-482M0QMA856:~$ ./a.out
Parent Process
Parent process id is 364
Child Process
Child process id is 2375
nithya@WIN-482M0QMA856:~$ |

```

Experiment:

```

nithya@WIN-482M0QMA856:~$ sleep 100 &
[2] 2548
nithya@WIN-482M0QMA856:~$ ps -p 2548 -o pid,stat,cmd
  PID STAT CMD
  2548 S    sleep 100
nithya@WIN-482M0QMA856:~$ cat /proc/2548/status
Name:      sleep
Umask:     0022
State:     S (sleeping)
Tgid:      2548
Ngid:      0
Pid:       2548
PPid:      364
TracerPid: 0
Uid:       1000    1000    1000    1000
Gid:       1000    1000    1000    1000
FDSize:    256
Groups:    4 24 27 30 46 100 1000
NSTgid:    2548
NSpid:     2548
NSpgid:    2548
NSSid:     364
Kthread:   0
VmPeak:    16112 kB
VmSize:    16112 kB
VmLck:     0 kB
VmPin:     0 kB
VmHWM:     7784 kB
VmRSS:     7784 kB
RssAnon:   1224 kB
RssFile:   6560 kB
RssShmem:  0 kB
VmData:    288 kB
VmStk:     132 kB
VmExe:     4832 kB
VmLib:     3240 kB
VmPTE:     72 kB
VmSwap:    0 kB
HugetlbPages: 0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads:   1

```

